

SHIVAM KUMAR PANDEY

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🔗 GitHub Portfolio Website Codeforces

Education

Indian Institute of Information Technology, Vadodara

December 2021 – July 2025

Bachelor of Technology in Computer Science and Engineering

Gujarat, India

Technical Skills

- **Languages:** Java, Kotlin, JavaScript, TypeScript, Python, C#
- **Frontend:** React, React Native, Tailwind CSS
- **Backend:** Spring Boot, Spring Security, Node.js, Express.js, REST API Web services, JWT, Microservices
- **Testing:** JUnit, Jest, Mocha, Unit Testing, Test Driven Development
- **AI/ML:** OpenAI API, Gemini API, Machine Learning, LLMs, RAG, Computer Vision, OpenCV, TensorFlow
- **Databases:** MongoDB, PostgreSQL, MySQL, Firebase Firestore
- **Cloud:** AWS Services(EC2, S3, Lambda, ECR, Bedrock), Docker, DevOps, CI/CD, Git
- **Problem Solving Skills:** DSA, OOD, Design Patterns, SDLC, Agile Development, Concurrency, System Design, Scrum, Distributed Systems

Work Experience

Rayole Software

Delhi

Software Engineer 1

June 2025 – Present

- Developed and published **12 scalable Mobile applications** using Agile development practices, showcasing excellent partnership skills with Design, Product and QA Testing teams through design systems, debugging, testing, and release cycles, serving **1.35M+ registered users** and **10k+ DAU**.
- Architected **5 Android Native plugins** in Java programming and implemented secure backend coding involving encrypted keys, nonce validation, postbacks, fraud detection, Google Play Integrity, and VPN detection.
- Integrated Firebase (Authentication, Firestore, FCM) and Singular MMP to enable real-time notifications, user tracking, and engagement optimization.
- Led integration of **4 AI/ML models**, including adaptive recommendation systems and Gemini LLM-powered workflows, improving user engagement and retention metrics (**D1 Retention: 25.81%**).
- Implemented multilingual support across **7 languages**, expanding product accessibility and user reach across multiple regions.

InnogenX

Remote

Software Engineer Intern

Jan 2025 – May 2025

- Designed and developed backend services for cloud-based applications using AWS, enabling secure asset management, scalable content delivery, and reducing application size from **700MB to below 70MB**.
- Integrated multilingual translation APIs supporting **3 languages** and implemented AWS Polly-based text-to-speech functionality for all supported languages.
- Implemented a virtual asset management environment leveraging local caching over AWS S3 storage(after first download), reducing application loading time by **3–5 seconds** per session.

Projects

Portfolio App | [Link]

Java, Android SDK, Resend API

- Developed an Android portfolio application featuring responsive UI, interactive gameplay, API-driven contact functionality, and modular application architecture.

SpendWise – Offline-First Expense Tracking Application

Kotlin, Jetpack Compose, MVVM, Hilt, Coroutines, StateFlow, Room, Firebase

- Built an offline-first Android application with **7+ screens** and **15+ user workflows**, supporting synchronization of **10,000+ expense records** between Room Database and Firebase Firestore.
- Implemented MVVM architecture, Hilt Dependency Injection, Coroutines, and StateFlow, reducing UI blocking operations by **90%** while maintaining smooth performance on low and mid-range devices.
- Implemented Repository Pattern, offline-first synchronization, and clean architecture principles to ensure reliable data consistency between local and cloud storage.
- Developed spending analytics, budget monitoring, and automated WorkManager notifications, achieving **sub-1.5 second startup time** and improving user engagement.

Explainable Visual Risk Analysis System for Construction Safety

Python, YOLOv8, Computer Vision, OpenCV, Explainable AI, React.js, Node.js, Express.js

- Built an explainable computer vision system using a YOLO-based object detection model to extract visual entities, followed by geometry-based reasoning and rule-driven risk management/assessment which lead to achieving **high accuracy of 85%**.
- Developed a full-stack demo application using Node.js, Express, and React to handle image uploads and visually present detection results and risk explanations. The overall workflow **response time is around 2-3 seconds**.
- Integrated an LLM(extraction and analysis on Big Data) strictly for post-inference explanation generation from structured risk data, ensuring explainability without hallucinated visual reasoning.

Achievements

- Research paper titled **On Enhancing Electric Vehicle Ecosystem in Urban Scenario** accepted at **IEEE-IATMSI 2025** and awarded **2nd Best Paper Award** among conference submissions.